

The One-Phase Stefan Problem: A Closed Neumann Root, Lambert-W Endpoints, and an Exact Energy Ledger

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Abstract

We give a source-local theorem for the dimensionless one-phase Stefan problem. The Neumann similarity ansatz reduces the moving boundary to the strictly increasing equation $F(\lambda) = \sqrt{\pi}\lambda e^{\lambda^2} \operatorname{erf} \lambda = \operatorname{Ste}$. This proves a unique positive root, supplies a five-term small-Ste inverse series and a two-sided large-Ste Lambert- W enclosure, and closes the wall/interface flux partition. Integrating the heat equation over the moving interval gives an exact sensible-plus-latent energy ledger. Zero superheat, zero diffusivity, and zero latent heat are labelled singular rescalings. The result is hydrodynamic/free-boundary analysis, not an arithmetic determinant or a Hilbert–Pólya construction.

1 Frozen model and similarity reduction

We scale the wall superheat, melting temperature, and thermal diffusivity to one. The liquid occupies $0 < x < s(t)$ and satisfies

$$u_t = u_{xx}, \quad u(0, t) = 1, \quad u(s(t), t) = 0, \quad \beta s'(t) = -u_x(s(t)^-, t), \quad s(0) = 0, \quad (1)$$

where $\beta = \operatorname{Ste}^{-1} > 0$ is the inverse Stefan number. With $\eta = x/(2\sqrt{t})$, the Neumann form is

$$s(t) = 2\lambda\sqrt{t}, \quad u(x, t) = 1 - \frac{\operatorname{erf}(\eta)}{\operatorname{erf}(\lambda)}. \quad (2)$$

The interface derivative is $-u_x(s(t)^-, t) = e^{-\lambda^2}/(\sqrt{\pi t} \operatorname{erf} \lambda)$. Consequently the Stefan condition is equivalent to

$$F(\lambda) := \sqrt{\pi}\lambda e^{\lambda^2} \operatorname{erf} \lambda = \operatorname{Ste}. \quad (3)$$

Theorem 1 (unique Neumann root and endpoint atlas). *For every $\operatorname{Ste} > 0$, equation (3) has exactly one $\lambda > 0$. Indeed, $F(0) = 0$, $F(\lambda) \rightarrow \infty$, and*

$$F'(\lambda) = \sqrt{\pi}e^{\lambda^2} \operatorname{erf} \lambda(1 + 2\lambda^2) + 2\lambda > 0. \quad (4)$$

The corresponding profile (2) solves (1) for $t > 0$. Writing $z = \operatorname{Ste}$, its small- z inverse is

$$\lambda^2 = \frac{z}{2} - \frac{z^2}{6} + \frac{7z^3}{90} - \frac{79z^4}{1890} + \frac{689z^5}{28350} + O(z^6), \quad (5)$$

and, for $z > 1$,

$$\frac{1}{2}W\left(\frac{2z^2}{\pi}\right) < \lambda^2 < \frac{1}{2}W\left(\frac{2(z+1)^2}{\pi}\right). \quad (6)$$

Thus $\lambda^2 \sim \frac{1}{2}W(2z^2/\pi)$ as $z \rightarrow \infty$.

Proof. Substitution of (2) into the heat equation uses $\frac{d}{d\eta} \operatorname{erf} \eta = 2e^{-\eta^2}/\sqrt{\pi}$; the two Dirichlet values are immediate. Differentiating F gives (4), while its endpoint limits follow from the standard limits of erf . Expanding

$$F(\sqrt{x}) = 2x + \frac{4}{3}x^2 + \frac{8}{15}x^3 + \frac{16}{105}x^4 + \frac{32}{945}x^5 + O(x^6)$$

and formal reversion gives (5). Finally, for $\lambda > 0$, $0 < \operatorname{erfc} \lambda < e^{-\lambda^2}/(\sqrt{\pi}\lambda)$, so

$$z < \sqrt{\pi\lambda^2}e^{\lambda^2} < z + 1.$$

The function $x \mapsto \sqrt{\pi x}e^x$ is increasing, and inversion with the principal Lambert branch proves (6). $\square$

2 Flux partition and the moving-domain energy identity

The wall and interface fluxes have a common $t^{-1/2}$ singularity:

$$J_{\text{wall}}(t) = -u_x(0, t) = \frac{1}{\sqrt{\pi t} \operatorname{erf} \lambda}, \quad J_{\text{int}}(t) = -u_x(s(t)^-, t) = e^{-\lambda^2} J_{\text{wall}}(t). \quad (7)$$

Thus $J_{\text{int}}/J_{\text{wall}} = e^{-\lambda^2}$ exactly. Integrating $u_t = u_{xx}$ on the moving interval (the boundary value $u(s(t), t) = 0$ removes the Reynolds endpoint term) gives

$$\int_0^t J_{\text{wall}}(\tau) d\tau = \int_0^{s(t)} u(x, t) dx + \beta s(t). \quad (8)$$

The elementary antiderivative $\int_0^\lambda \operatorname{erf} \eta d\eta = \lambda \operatorname{erf} \lambda + (e^{-\lambda^2} - 1)/\sqrt{\pi}$ therefore yields

$$S(t) := \int_0^{s(t)} u dx = \frac{2\sqrt{t}(1 - e^{-\lambda^2})}{\sqrt{\pi} \operatorname{erf} \lambda}, \quad (9)$$

$$L(t) := \beta s(t) = 2\beta\lambda\sqrt{t} = \frac{2e^{-\lambda^2}\sqrt{t}}{\sqrt{\pi} \operatorname{erf} \lambda}, \quad (10)$$

$$I(t) := \int_0^t J_{\text{wall}}(\tau) d\tau = \frac{2\sqrt{t}}{\sqrt{\pi} \operatorname{erf} \lambda} = S(t) + L(t). \quad (11)$$

This is an exact sensible-plus-latent energy ledger, not a fitted numerical balance. The eight rational probes in the accompanying receipt cover both terms and the Lambert enclosure.

Source note

The classical free-boundary setting and small-latent-heat asymptotics are documented by Addison, Howison, and King [1]; phase-change and Neumann conventions are treated in Gupta [2]; two-phase stability background is given by Rubinstein [3]. These references are source metadata, not a priority or arithmetic claim.

References

- [1] J. A. Addison, S. D. Howison, and J. R. King, *Ray methods for Free Boundary Problems*, Oxford author manuscript (2005), [smallstefan.pdf](#).
- [2] S. C. Gupta, *The Classical Stefan Problem: Basic Concepts, Modelling and Analysis*, Elsevier, North-Holland Series in Applied Mathematics and Mechanics, Volume 45 (2003), ISBN 978-0-444-51086-0.
- [3] L. I. Rubinstein, Global Stability of the Neumann Solution of the Two-phase Stefan Problem, *IMA Journal of Applied Mathematics* 28(3), 287–299 (1982). DOI: 10.1093/imamat/28.3.287.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic or operator claim. **Data and code.** The theorem, ledger, independent checks, and build instructions are released with HCS-C226. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked displayed claims and metadata. This is not external peer review.